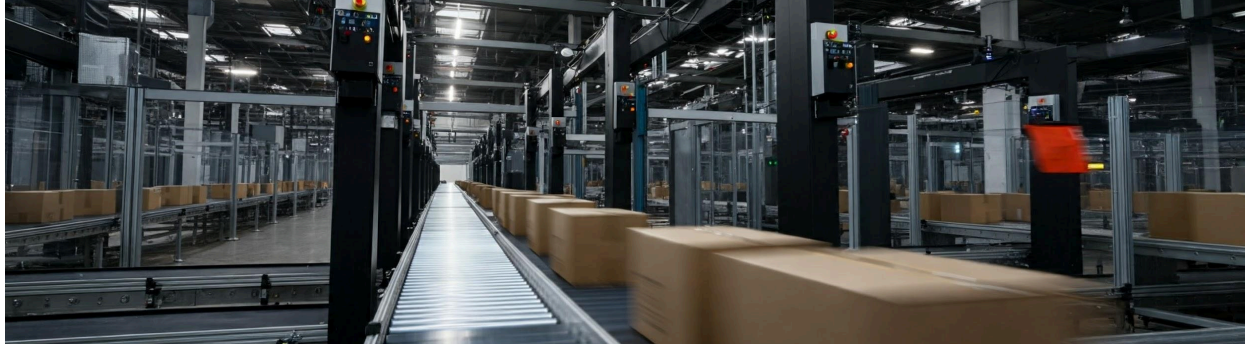




RFID requirements for Distributors managing On's products in the warehouses

Updated 07.07.2026 adding a Q&A section, changing the POC and updating the manual outbound process guidelines

GS1 Standard:

The standard we follow for the RFID encoding is:

<https://www.gs1.org/services/epc-encoderdecoder>

Current tags on the products:

- Shoes: SL3S1206 Chip: NXP UCODE 9, Inlay AZ-H91 Antenna Dimensions 70*14
- Apparel: AD-387 Chip:NXP UCODE 9 Antenna Dimensions 50 x 30 mm / 1.97 x 1.18 in
- Accessories: AD-327 U9 ETSI, Chip NXP UCODE 9 Antenna Dimensions 41 x 16 mm / 1.63 x 0.63 in

Outbound process

The outbound process is a mandatory requirement for all distributors handling On products. Please note that there are two methods available for transmitting this outbound information, which are detailed in the subsequent sections of this document.

Option A: Automated Outbound process with API connection

It is a mandatory requirement to share the outbound data with On with the RFID granularity and the order information:

- Based on the distributor's process - the hardware is connected to the WMS (usually with Carton Id or Order number)
- As On, we need to have Point of sales name, address, city, and the RFID tags info (24 digit code)

This requirement is valid for all the On products leaving the distributor warehouse

Data structure for Outbound

1) WMS to NEDAP

url for test:

<https://nedap-harmony.com/api/fefb753e-3e56-418b-89c5-8778e79c4aec/hooks/v1/distributor-hook/jobs/run?test=true>

url for production:

<https://nedap-harmony.com/api/fefb753e-3e56-418b-89c5-8778e79c4aec/hooks/v1/distributor-hook/jobs/run>

For more information about the url:

<https://documentation.nedap-harmony.com/reference/runhook>

Body of the Post Call:

```
{
  "distributorName": "XXXXX",
  "orderType": "B2B",
  "boxReference/Order number": "YYYY",
  "destination": {
    "name": "Company XYZ",
    "city": "XX",
    "street": "YY",
    "country": "ZZ"
  },
  "boxes": [
    {
      "boxReference": "YYYY",
      "epcs": [
```

```

"3035D0D0C408AC000098C35B",
"3035D0D0C408AC000098C35E"
],
},
{
  "boxReference":"QQQ",
  "epcs":[
    "XXXXXXXXXXXXXXXXXXXXXXXXXXXX",
    "XXXXXXXXXXXXXXXXXXXXXXXXXXXX "
  ],
  }}
}

```

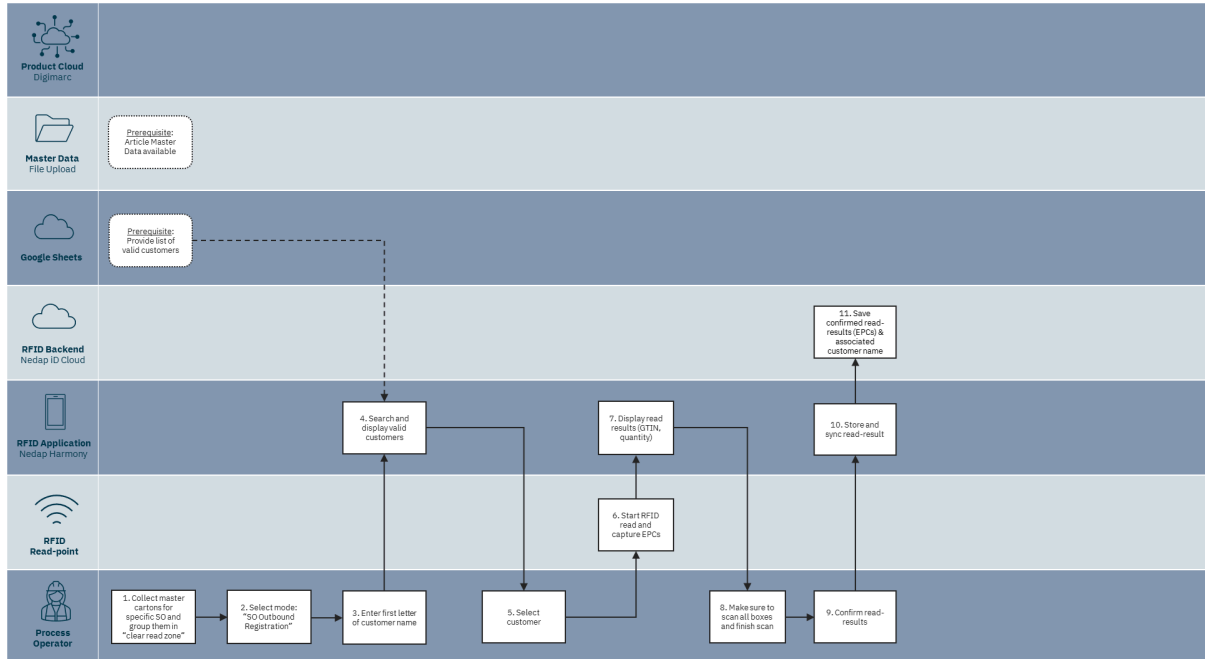
Option B: Manual Outbound process with Nedap Harmony

This document outlines the necessary steps to onboard the On's distributors who receive direct shipments from the factories for the RFID project.

STEP 1:

The sales contact person for the distributor will:

1. Inform the distributor about the project and set a timeline for the onboarding
2. Share the recommended devices needed to perform the RFID scanning
 - a. Handheld devices supported (cost estimation 2-3K USD) [Handhelds](#)
 - i. Preferable device: Zebra RFD4030 (Sled) TC21/26, EC50/55 (mobile) version with 8-pin connector (eConnex)
 - b. Fix readers devices supported (cost estimation 2-3K USD) <https://documentation.nedap-harmony.com/docs/fixed-readers>
3. Share with the distributor the process flow needed to implement
 - a. Outbound scanning + customer identification



4. The distributor will have available a list of the customers (POS) to share with On organised with the following information

Distributor name	Account Name	POS Account Name	POS City	POS Street
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Please note: a new system is being roll-out, which will replace the google sheet, and give the distributors access to salesforce directly.

STEP 2:

Once the distributor has purchased the device and agreed on the process flow, the distributor needs to make sure they have:

- The device (handheld reader) available on hand
- Access to Google Play Store from the device
- A Google account linked with the device
- Stable Wifi connection
- Some On's products on hand for testing purposes
- Filled in the POS list with all your current customers
- People from the operation, who will be in charge of performing the RFID scanning on a daily basis

After having all of this ready, these are the tasks that come after:

- The sales contact person from On will:

- Create the tab for the distributor in the [POS list](#)
- Create a new Read Point via the internal ticketing system and once this is created, add a Touch Point in Nedap and share this information with the distributor.
- Share the [training guides](#) and the [support procedure](#)
- The distributor will:
 - Configure and test the device with the following [setup guide](#) (from step 9 onwards) based on the read point and touch point information that was shared by the sales contact person from On
 - Share the point of contact for the RFID solution with the sales contact person from On
 - Go live with the solution once testing has been done and the data has been validated.

STEP 3:

The sales contact person for the distributor can opt to receive periodic reports from Looker, which visualize the list of shipped products and quantities per point of sale.

Frequent Questions & Answers

API Integration & Outbound Data

Q: Can a sample EPC and UPC be provided for validation purposes?

A: Sample HEX EPC: [302ED64632F03133394221CE](#), which corresponds to URI [urn:epc:id:sgtin:7615537.071862.10000213](#).

Q: Does the EPC UPC format align with our WMS configuration?

A: The encoding follows standard GS1 GTIN-12 / UPC-A. The distributor should confirm internal WMS compatibility.

Q: Is there a guide for decoding the EPC (e.g., "ad72120643eb1fb1867c005c")?

A: HEX EPCs can be translated into URIs using the official [GS1 EPC Encoder/Decoder](#). This identifies the Company Prefix, Item Reference, and Serial Number.

Q: What are the Nedap account and connection details (UAT/Prod) for integration?

A: The sales contact person from On will contact Nedap (support@nedap-harmony.com) and add the distributor in CC to request the site-specific API connections for UAT and Production for the Outbound Process.

Q: Will there be a specific partner or franchisee code for the distributor?

A: Yes, the sales contact person will assign a certain name for the distributor that will be used in the message transmitted via API in the "*distributorName*": "XXXXX" section.

Q: What scenarios are required to follow the RFID requirements?

A: It is needed for all outbound processes, including Replenishment / Stock Transport Orders (STO) and Outbound Delivery for a Sales Order.

Q: Is there a limit to the number of EPCs in the message transmitted via API?

A: Technically, the API accepts a JSON array of string values or a comma-separated string, so there is no hard-coded limit. However, we recommend batching EPCs (e.g., 5,000 to 10,000 per payload) to ensure optimal performance.

Q: Is there a character restriction in the message transmitted via API for the fields name, city, street and country?

A: While the backend can accept long strings, we recommend a standard maximum of 50 to 100 characters to ensure the text displays correctly in the Nedap Harmony UI.

Nedap Harmony System & Hardware Setup

Q: What is the distinction between "Account Name" and "POS Account Name" in the POS Master document?

A: The Account Name identifies the distributor organization, whereas the POS Account Name specifies the individual stores or selling points associated with that distributor which might be located in different cities.

Q: What are the minimum local network requirements for deploying Nedap Harmony?

A: The local store network must have a DHCP server to assign IP addresses automatically to fixed readers. Ensure outbound traffic is unrestricted on Port 443 (HTTPS) for general communication and Port 8883 (MQTTS) for secure, real-time message queuing to the Nedap IdCloud endpoints. Wi-Fi networks must maintain a signal strength of at least -65 dBm with low latency across all retail areas.

Q: Does Nedap Harmony require any local servers or on-premise databases?

A: No. Nedap Harmony is a fully cloud-native SaaS platform built on the Nedap IdCloud infrastructure. All data storage, product lookup databases, and reporting engines are hosted securely in the cloud. Hardware and mobile applications communicate directly with the cloud endpoints, eliminating the need for on-premise server hardware.

Q: How do we configure local firewalls to allow Nedap devices to connect?

A: Distributors should whitelist the wildcard domain *.idcloud.nedap.com on the store's firewall. If IP whitelisting is strictly required by the IT department, contact Nedap Support to receive the current list of regional cloud gateway IP addresses, though domain-based whitelisting is highly recommended due to dynamic cloud scaling.

Q: How does a distributor initialize the Nedap IdCloud app on a new store mobile device?

A: 1. Install the "Nedap IdCloud" app from the official Google Play Store or iOS App Store, or push it via your MDM platform. 2. Log in to the Nedap Harmony Portal, navigate to the store configuration tab, and generate a Store Provisioning QR Code. 3. Open the app on the mobile device and tap "Scan Provisioning Code". 4. Point the camera at the QR code. The app will automatically download all store-specific metadata, configure server settings, and apply local security policies.

Q: Can a single mobile device be shared between multiple physical store locations?

A: Yes, but it must be re-provisioned. To switch stores, the operator must log out, clear the local app cache, and scan the Provisioning QR Code of the new store. To ensure data integrity, it is highly recommended to dedicate specific devices to a single store location.

Q: How do we verify that the mobile app is successfully communicating with the cloud?

A: In the IdCloud app, check the cloud icon on the top navigation bar. A solid green cloud icon indicates a live, stable connection. A yellow or red icon with a strike-through indicates a local offline state or blocked outbound ports on the Wi-Fi network.

Q: What format is required for importing product master data into the system?

A: Product master data must be uploaded to the Nedap Harmony portal via secure SFTP or REST API in a standardized CSV or JSON format. Each record must contain the following mandatory fields: GTIN/EAN, SKU, Brand, Product Name, Color, Size, and Department. The system uses this file to translate scanned EPC codes into consumer-readable product details.

Q: How does the system map physical inventory to specific zones like "Sales Floor" vs. "Stockroom"?

A: The Nedap Harmony system is built around defined "sub-locations" or "zones". Before starting an inventory scan, the operator selects their current zone within the app (e.g., "Scanning Sales Floor"). Any RFID tags registered during that session are automatically mapped to that location in the database. For automated fixed readers (iD Top), the transition direction (e.g., leaving the stockroom) determines the automatic zone update.

Q: What standard of RFID tags are compatible with Nedap Harmony?

A: The system is fully compatible with standard EPC Gen2 (ISO 18000-6C) UHF RFID tags. It supports standard GS1 encoding schemes, including SGTIN-96 and GRAI-96. Ensure your label supplier programs the tags according to these official global standards.

Q: What is "RFID Bleed-through" and how do we calibrate against it?

A: Bleed-through occurs when a reader set to high power scans tags through walls or ceilings (e.g., reading stockroom items while standing in the sales floor). To calibrate, use the Power Attenuation slider inside the IdCloud App's diagnostic menu. Reduce the power level progressively until the scanner only registers tags within a 2-to-3-meter radius, ensuring clean separation between store zones.

Q: The mobile app displays "No Scanner Connected". How do we resolve this?

A: 1. Verify that Bluetooth is enabled on the mobile device. 2. Ensure the iD Hand scanner is turned on and the battery is charged (green battery LED). 3. If previously paired to another device, turn off Bluetooth on that device to release the connection. 4. Perform a hard reset of the iD Hand by holding the power button for 10 seconds, then attempt the pairing process again.

Q: How do we retrieve hardware system logs for advanced support diagnostics?

A: Connect the fixed reader or mobile device to a local PC on the same subnet. Access the reader's local web console by typing its IP address into a browser. Log in with your authorized distributor credentials, navigate to the "System Logs" tab, and click "Export Diagnostic Package" to download a zip file containing detailed logs for Nedap Tier-3 engineering.